CMAN :=

Given $N = 1024$, $f = 50\,Hz$

Task 1:=

$R = N \times f = 1024 \times 50 = 51{,}200$ spikes/second

Task 2:=

$B = R \times 20$ (bits per packet total)

$B = 51{,}200 \times 20 = 1{,}024{,}000$ bits/ seconds

$B = 1.024$ Mbits/s

Task 3 :=

Given, SPI (≤ 50 Mbit/s), I^2C (≤ 3.4 Mbit/s),
AXI4-Lite (assume 100 Mbit/s)

from task 2, $B = 1.024$ Mbits/s

$max \geq B$

$I^2C \Rightarrow 3.4\,Mbit/s \geq 1.024\,Mbit/s$

$SPI \Rightarrow 50\,Mbit/s \geq 1.024\,Mbit/s$

$AXI4 \Rightarrow 100\,Mbit/s \geq 1.024\,Mbit/s$

so, I^2C is the lowest complexity than SPI and AXI4 - lite.

Task 4:=

25% of 1024 neurons fire at once

$$\frac{25}{100} \times 1024 = 0.25 \times 1024$$

Neuron firing = 256 Neurons

Each neuron that fire sends 1 Packet = 20 bits
(From task 2)

Total bits = Neuron firing × bits per packet

= 256 × 20 = 5,120 bits

Total bits = 5120 bits.

1 ms = $\frac{1}{1000}$ second = 0.001 seconds

Peak bandwidth = $\frac{5120}{0.001}$ = 512000 bit/s

From task 2, Bandwidth = 5.12 M bit/s

Bmean = 1.024 M bits/s

Burst Bandwidth (Bburst) = 5.12 Mbit/s

$$\text{Ratio} = \frac{B_{burst}}{B_{mean}} = \frac{5.12}{1.024} \approx 5\times$$

From Task 3, we chose I^2C

3.4 > 5.12, No, its too slow.

Here we have multiple choices

Choice A : We can Add a buffer.

Buffer depth = Burst Packets = 256 Packets

Choice B :: Upgrade to SPI

50 ≥ 5.12, it handles the burst easily.

No buffer is needed in this SPI.

We have a couple of option

option ① :: choose I^2C, it's too slow.

we can add a buffer.

overflow $= 256 \times 20 - 3400$

$= 5120 - 3400 = 1720$ bits

in terms of packets.

$\frac{1720}{20} = 86$ packets.

so, roughly the size of the buffer would be 256 bytes.

option ② To go with the SPI (upgrade)

$50 \geq 5.12$ (it handles the burst easily)

No buffer is needed in this SPI.

Task 5:=

Given, All 1024 neurons and sends 1bit per neuron fired.

Bits per frame = 1024 neurons x 1 bit per neuron

= 1024 bits per frame

$$\text{Frame per second} = \frac{1\,\text{second}}{1\,\text{ms}} = \frac{1}{0.001} = 1000\ \text{frames/second}$$

B_{frame} = Bits per frame x frames per second

$= 1024 \times 1000 = 1024000$ bit/s

$B_{frame} = 1.024$ M bit/s

From task 2:= $B_{AER} = 1.024$ Mbit/s

$$\text{Ratio} = \frac{B_{AER}}{B_{frame}} = \frac{\cancel{1.024M}\ \text{bit/s}}{\cancel{1.024M}\ \text{bit/s}}$$

Ratio = 1

$B_{AER} = B_{frame}$

$\cancel{N} \times f \times 20 = \cancel{N} \times 1000 \times 1$

$f \times 20 = 1000 \times 1$

$$f = \frac{\cancel{1000}^{50}}{\cancel{20}} = 50$$

$$\boxed{f_{crossover} = 50\ \text{Hz}}$$

$f < 50 HZ$ $\leftrightarrow$ $f = 50 HZ$ $\leftrightarrow$ $f > 50 HZ$

AER wins — equal — frame wins

$B_{AER} < B_{frame}$, Because AER uses less Bandwidth because most neurons are quiet, no point sending empty data.

When neurons fire fast ($f > 50 HZ$)

$$B_{AER} > B_{frame}$$

AER uses more bandwidth because every packet has 20 bits of overhead but frame based only uses 1 bit per neuron.

So, AER is the right choice when your neurons are mostly quiet because why waste bandwidth reporting nothing happened for hundreds of neuron when we can just send a small packet only when something actually fires.